

HACK MATRIX HACKATHON

TITLE PAGE

- **PROJECT TITLE :** **BioDex - Gamified Urban Wildlife Tracker**

Under Wildlife Sighting Reporter Problem ID: HM44

- **TEAM NAME :** **Debug Duo**

- **TEAM MEMBERS :** **Prateek Das**
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PROBLEM STATEMENT

- **Describe the problem clearly .** Urban biodiversity data is incredibly scarce because everyday citizens lack a simple, structured, and engaging way to contribute local wildlife observations.
- **Who is affected by this problem .** Environmental researchers tracking urban ecosystems, and local communities disconnected from their surrounding wildlife.
- **Why this problem is important ?** Without crowdsourced data, it is impossible to monitor species decline, invasive species spread, or the general ecological health of our immediate urban environments.

- **List of existing Solutions (if any)**

Complex academic portals (like eBird) and unstructured posts in local community WhatsApp/Facebook groups.
- **Mention their Limitations**

Academic tools have steep learning curves that deter casual users. Social media posts lack standardized GPS tagging and searchable databases.
- **Explain the gap your project will address**

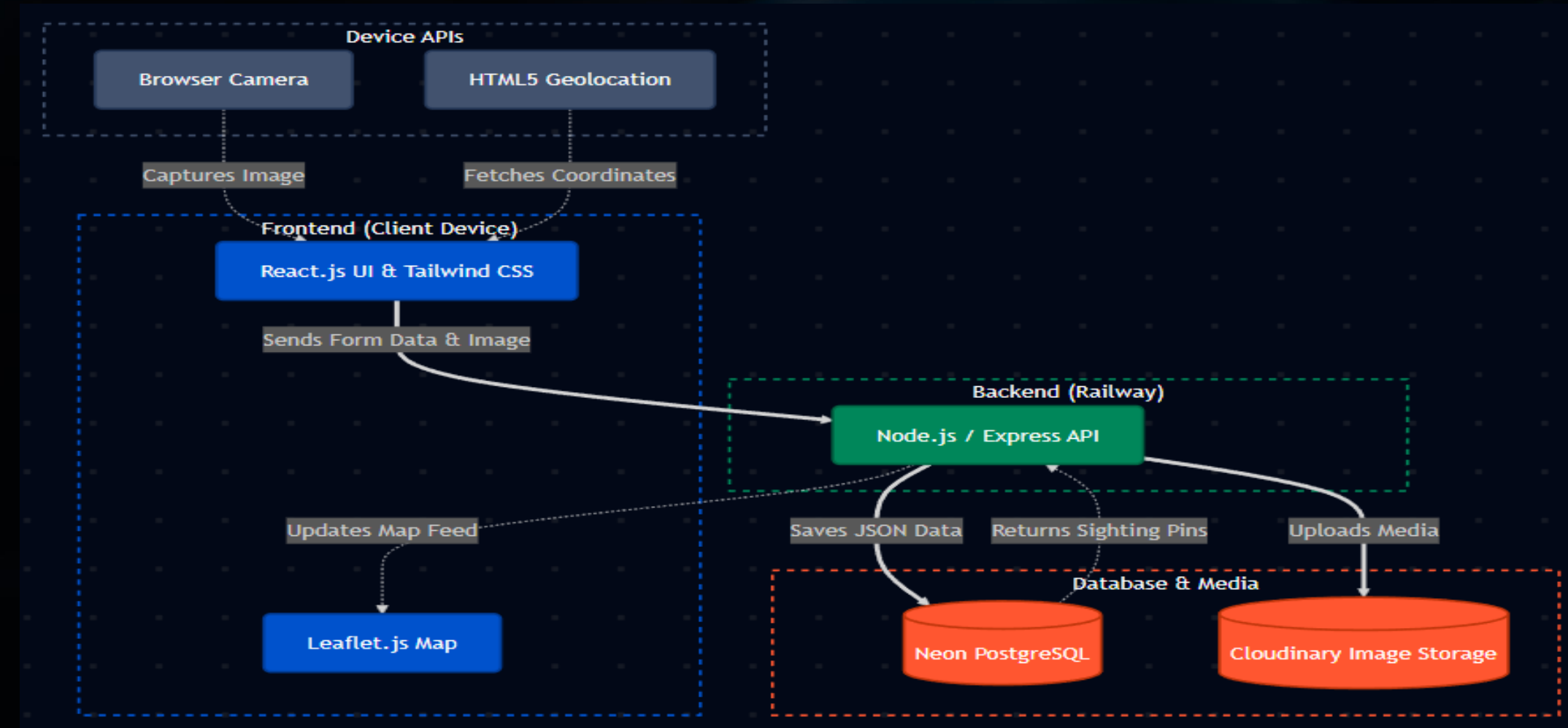
BioDex bridges the gap by turning ecological data collection into an engaging, gamified experience for the everyday citizen, automatically capturing standard GPS and timestamp data without extra effort.

PROPOSED SOLUTION

- **Explain your solution clearly .** A mobile-responsive web app where users photograph local wildlife, log it with automatic GPS coordinates, and view community sightings on an interactive map.
- **List key features** A photo-first sighting submission form, an interactive real-time community map, and a gamified "Species Explorer" badge system.
- **Explain how it solves the problem** It removes the friction of data entry and introduces a "collect-them-all" motivation, seamlessly generating rich, localized biodiversity data.

SYSTEM ARCHITECTURE

- Include architecture diagram

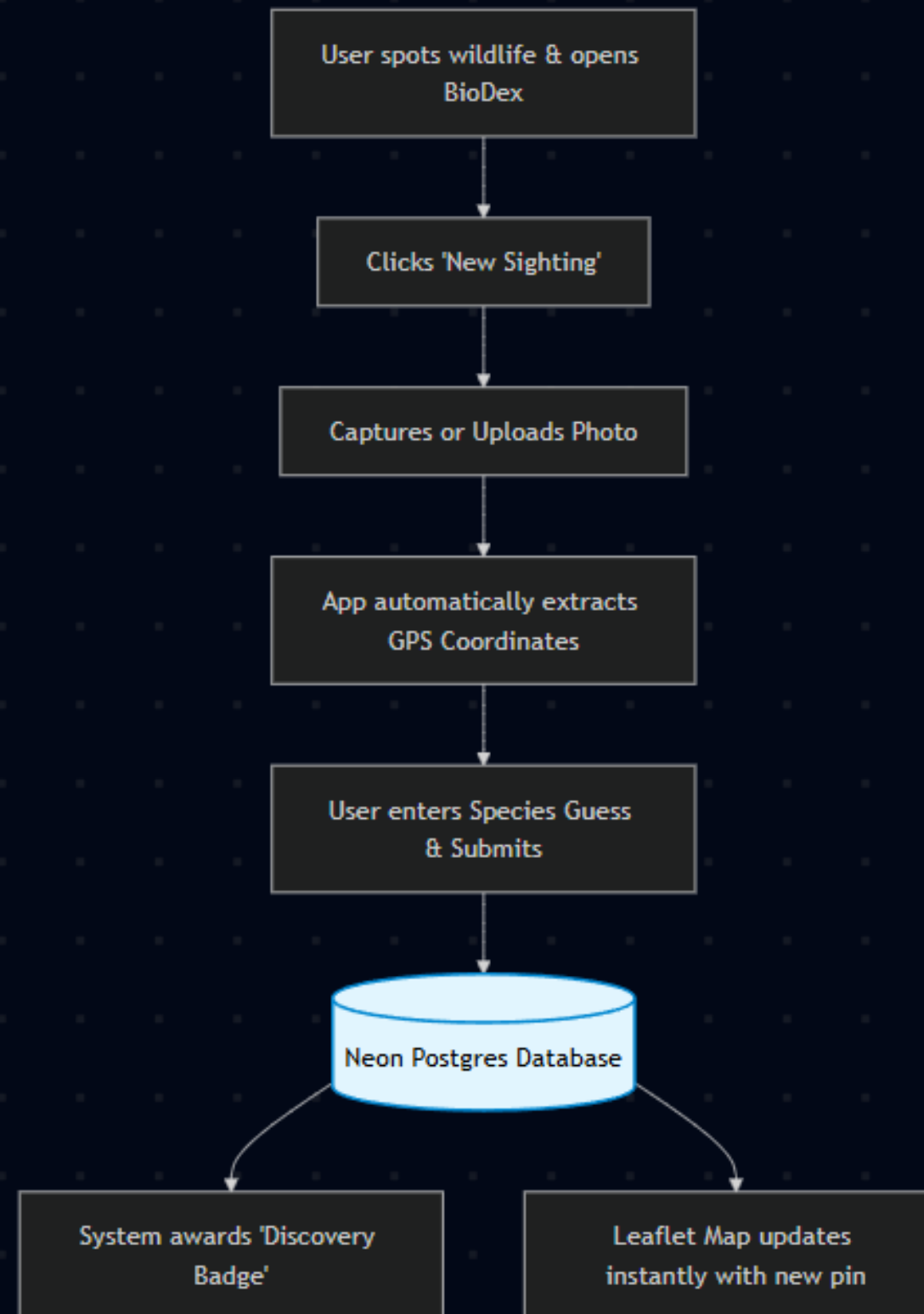


- Mention main components
(Frontend, Backend, Database, APIs)
 - Frontend: React.js (User Interface)
 - Backend: Node.js / Express.js (Hosted on Railway)
 - Database: PostgreSQL via Neon Tech (Relational Data) & Cloudinary (Image Storage)
 - APIs: HTML5 Geolocation API, Leaflet.js

- Explain data flow briefly

The React client captures location and photo -> Photo goes to Cloudinary -> Backend receives Image URL + GPS data -> Saves to Neon Postgres Database -> Pushes data back to update the Leaflet map.

- Show step-by-step working
- Include user flow or system flow
- Use a proper flowchart



TECH STACK

- **Frontend technologies** React (Vite) and Tailwind CSS - For rapid, responsive mobile-first UI development.
- **Backend technologies** Node.js with Express (Hosted on Railway) - For robust REST API creation and fast CI/CD deployment.
- **Database** Neon Tech (Serverless PostgreSQL) for sighting data, and Cloudinary for wildlife image storage.
- **Tools and APIs used and reason for choosing each (1 line each)**
 - Leaflet.js: Lightweight, open-source mapping without requiring API keys.
 - HTML5 Geolocation API: Native browser support to easily capture accurate coordinates.

IMPLEMENTATION PLAN

- **Step-by-step development plan**

- 1) Scaffold UI and initialize Map.
- 2) Build photo upload and GPS capture logic.
- 3) Connect to Firebase.
- 4) Render dynamic pins on the map.

- **Modules or features to be built**

Interactive Map View, Sighting Submission Form, Gamification/Badge Engine.

- **Timeline (if applicable)**

- 9:30 AM - 12:30 PM: Core React UI & Map setup.
- 12:30 PM - 3:30 PM: Form logic & Geolocation API.
- 3:30 PM - 5:30 PM: Styling and Testing

EXPECTED OUTCOME / IMPACT

- **Expected results of the project**

A functional prototype featuring an interactive map populated with wildlife pins and a working submission form capturing real location data.
- **Benefits to users**

Citizens get a fun, engaging way to interact with nature, learn about local species, and earn digital rewards.
- **Real-world impact**

Creates a valuable, crowdsourced dataset of urban biodiversity that can be exported for environmental research and local conservation planning.

- **Future improvements**

Integrating an AI Vision model to automatically identify the animal in the uploaded photo, reducing user entry error.
- **Final summary**

Expanding to a nationwide platform with regional leaderboards and partnerships with schools for environmental education.
- **Scalability of the project**

BioDex proves that by combining accessible mapping tech with gamified user experiences, we can solve the urban biodiversity data shortage one photo at a time.